

ClonEval: An Open Voice Cloning Benchmark

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Abstract—We present a new benchmark for voice cloning text-to-speech models. The benchmark consists of an evaluation protocol, an open-source library for assessing the performance of voice cloning models, and an accompanying leaderboard. The paper discusses design considerations and presents a detailed description of the evaluation procedure. The usage of the software library is explained, along with the organization of the leaderboard. The evaluation results of selected open-source models are reported.

Index Terms—voice cloning, speech synthesis, evaluation

I. INTRODUCTION

RECENT speech synthesis models, such as VALL-E [1] and Voicebox [2], can clone arbitrary voices. Many open-source alternatives have also become available. These developments create an increasing demand for reproducible methods to evaluate voice cloning capabilities. The reliable procedure for evaluating voice cloning should not assume any particular architecture of the model under study. It should also enable assessment of models that are not publicly available without excessive effort. To make the procedure reproducible it should have a form of a standardized benchmark with reasonable defaults set for data being utilized and evaluation steps being conducted. In order to facilitate the comparison of models with each other the benchmark should adopt the measurement techniques that are already accepted in the field while filling any under-specified parameters that could impact the evaluation result in an unpredictable way.

To address the aforementioned needs, we propose *ClonEval*, a reproducible benchmark for voice cloning that consists of:

- 1) A deterministic evaluation protocol that sets defaults for data, metrics, and models to be used in the voice cloning assessment process.
- 2) An open-source software library that can be used to evaluate voice cloning models in a reproducible manner.
- 3) A public leaderboard that enables comparison of the models against each other.

The benchmark utilizes speaker embeddings to measure similarity between reference and generated samples. Besides establishing ranking with overall similarity, we also analyze how well particular emotions are transferred by models under study to obtain a more fine-grained view on their performance in capturing nuances. For the purpose of illustrating the use of our benchmark, we determine and analyze scores for selected open-weight models that were available online at the time of writing.

II. RELATED WORK

Current methods of evaluating voice cloning models have significant limitations. Both [1] and [2] employed word error rate (WER) to evaluate the intelligibility of synthesized speech. However, this metric only captures text accuracy and does not offer insight into voice cloning quality. Additionally, [2] conducted a human evaluation and reported mean opinion score (MOS) for speaker quality and similarity. While informative, subjective assessments like these can be inconsistent and difficult to reproduce.

In order to evaluate speaker similarity directly, both [1] and [2] used the WavLM-TDNN model to obtain speaker embeddings for the prompt and the generated audio, and calculate the similarity between them. [2] also introduced the Fréchet Speech Distance (FSD), an adaptation of Fréchet Inception Distance (FID), which calculates distribution-level similarity between real and synthesized speech using features extracted with wav2vec 2.0.

Seamless Communication et al. [3] propose a new approach through AutoPCP which is a neural model designed to predict the human-rated prosody similarity (PCP score) between two audio signals. However, relying solely on neural networks for evaluation can introduce bias or misalignment with human perception.

III. BENCHMARK DESIGN

In order to develop a benchmark for voice cloning models, the following design principles were adopted:

- 1) The voice cloning models should be treated as opaque, rather than assuming any particular architecture of the proposed solutions.
- 2) There should be no need for human intervention in the evaluation process.
- 3) The evaluation procedure should be easily adaptable to new models.
- 4) The evaluation results should be reproducible.

As voice cloning technologies evolve rapidly, the evaluation procedure should be designed in a way that is resistant to technological changes that will appear in future text-to-speech (TTS) models. To fulfill principle (1), the requirements imposed on the evaluated model are reduced to an absolute minimum. The model must accept an audio sample of the voice to be cloned and a text prompt of an utterance as an input. The model is expected to generate a single audio sample as the

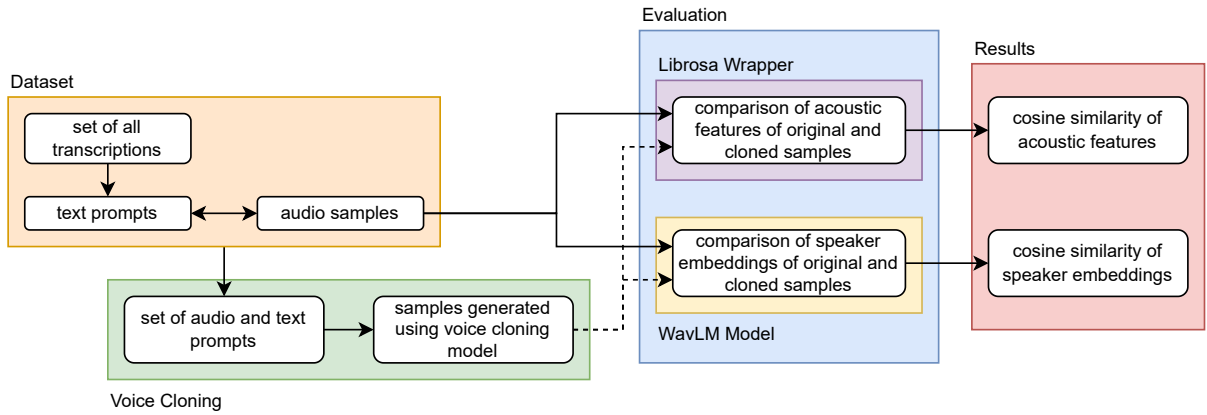


Fig. 1. Overview of the evaluation process.

output, representing the result of voice cloning. No additional assumptions, such as the ability to provide confidence scores or n -best lists, are made.

While determining the mean opinion score (MOS) is a widely accepted practice for the subjective evaluation of TTS quality, adopting this practice in the context of an open benchmark would present significant challenges. Establishing an arena for comparing different models, as presented by Chiang et al. [4], and collecting human preference judgements would require curating the list of available models for evaluation. Such an approach presents challenges, as it prevents independent researchers from evaluating models that are not publicly available due to commercial or ethical considerations. Relying on the results reported by the model authors could compromise the credibility of the findings, as verifying the accuracy of the reported results would require re-running the human evaluation, which is not feasible. Therefore, we have decided to adopt principle (2).

Following the previous approaches [1], [2], we assess the similarity between the original and cloned samples using a pretrained WavLM [5] model, which utilizes its learned universal speech representations to achieve effective performance in numerous speech processing tasks in English. To fulfill principles (3) and (4), we developed *ClonEval*, a software library that encapsulates the entire evaluation process. It can be used by vendors of voice cloning models to test their models independently and report their scores. The reported results can be reproduced with the use of the *ClonEval* library by an external entity that has no access to the model, if the vendor releases cloned samples.

IV. EVALUATION PROCEDURE

The evaluation procedure outlined in Figure 1 involves two stages. First, samples are generated using a voice cloning model, with the primary dataset for cloning being LibriSpeech test-clean [6]. This dataset is currently one of the most widely used for model evaluation in many English speech processing tasks. Four additional popular English emotional speech datasets were selected to investigate the impact of

emotionality on voice cloning quality. The selected emotional datasets were CREMA-D [7], RAVDESS [8], SAVEE [9], and TESS [10]. As previously mentioned, the model must take as input a sample of the voice to be cloned and a text sample of an utterance. To obtain the text samples, for each voice sample, one text prompt was drawn from all those available in the given dataset, excluding the prompt corresponding to that sample. Since the RAVDESS dataset contains only two distinct utterances in total, a single text prompt is used per evaluation sample. Under this constraint, randomly drawing one alternative prompt provides a consistent protocol across all datasets. It also ensures that the generated utterance does not repeat the text of the audio prompt.

Following the generation of samples through the voice cloning model, an evaluation is conducted using the WavLM model, with each sample undergoing identical processing. Given that the WavLM model accepts audio samples with a sampling rate of 16 kHz as input, the audio is resampled to this value. Speaker embeddings are then generated using the WavLM model.¹ For each pair of samples (reference and generated), the cosine similarity between their speaker embeddings from WavLM is calculated. Since cosine similarity is a standard choice in existing voice cloning evaluations based on speaker embeddings [1], [2], it is used as the primary metric. The similarity values obtained for all samples from a given dataset are averaged to obtain the final evaluation result.

For the purpose of conducting fine-grained error analysis, we also extract acoustic features from each sample with Librosa [11]. The extracted acoustic features and parameters used are shown in Table I.

V. SOFTWARE LIBRARY

The evaluation code and setup guidelines are provided in the GitHub repository². The input files for the evaluation must be organized into two directories: one containing reference samples and the other containing generated samples. Each directory should contain files with identical filenames for

¹[microsoft/wavlm-base-plus-sv](https://github.com/microsoft/wavlm-base-plus-sv)

²<https://github.com/amu-cai/cloneval>

TABLE I
ACOUSTIC FEATURES EXTRACTED FROM AUDIO SAMPLES.

Feature	Parameters
pitch	fmin = 65 Hz, fmax = 2093 Hz, sr = 16 kHz, frame_length = 2048, hop_length = 512, n_thresholds = 100, beta_parameters = (2,18), boltzmann_parameter = 2, resolution = 0.1, max_transition_rate = 35.92, switch_prob = 0.01, no_trough_prob = 0.01, fill_na = nan, center = True, pad_mode = constant
spectrogram	n_fft = 2048, hop_length = 512, win_length = 2048, window = hann, center = True, dtype = None, pad_mode = constant, out = None
Mel spectrogram	sr = 16 kHz, n_fft = 2048, hop_length = 512, win_length = 2048, window = hann, center = True, pad_mode = constant, power = 2.0
MFCCs	sr = 16 kHz, n_mfcc = 13, dct_type = 2, norm = ortho, lifter = 0, mel_norm = slaney
RMS	frame_length = 2048, hop_length = 512, center = True, pad_mode = constant, dtype = float32
spectral centroid	sr = 16 kHz, n_fft = 2048, hop_length = 512, freq = None, win_length = 2048, window = hann, center = True, pad_mode = constant
spectral bandwidth	sr = 16 kHz, n_fft = 2048, hop_length = 512, win_length = 2048, window = hann, center = True, pad_mode = constant, freq = None, centroid = None, norm = True, p = 2
spectral contrast	sr = 16 kHz, n_fft = 2048, hop_length = 512, win_length = 2048, window = hann, center = True, pad_mode = constant, freq = None, fmin = 200.0, n_bands = 6, quantile = 0.02, linear = False
spectral flatness	n_fft = 2048, hop_length = 512, win_length = 2048, window = hann, center = True, pad_mode = constant, amin = 1e-10, power = 2.0
spectral roll-off	sr = 16 kHz, n_fft = 2048, hop_length = 512, win_length = 2048, window = hann, center = True, pad_mode = constant, freq = None, roll_percent = 0.85
zero crossing rate	frame_length = 2048, hop_length = 512, center = True
LPCs	order = 2, axis = -1
tempogram	sr = 16 kHz, hop_length = 512, win_length = 384, center = True, window = hann, norm = inf, with the following onset strength envelope parameters: sr = 16 kHz, lag = 1, max_size = 1, ref = None, detrend = False, center = True, feature = librosa.feature.melspectrogram(fmax=sr/2), aggregate = np.mean
chromagram	sr = 16 kHz, n_fft = 2048, hop_length = 512, win_length = 2048, window = hann, center = True, pad_mode = constant, tuning = None, n_chroma = 12
pseudo-constant-Q transform	sr = 16 kHz, hop_length = 512, fmin = 32.7 Hz, n_bins = 84, bins_per_octave = 12, tuning = 0.0, filter_scale = 1, norm = 1, sparsity = 0.01, window = hann, scale = True, pad_mode = constant, dtype = None
IIRT	sr = 16 kHz, win_length = 2048, hop_length = 512, center = True, tuning = 0.0, pad_mode = constant, flayout = sos, res_type = soxr_hq
variable-Q transform	sr = 16 kHz, hop_length = 512, fmin = 32.7 Hz, n_bins = 84, intervals = equal, gamma = None, bins_per_octave = 12, tuning = 0.0, filter_scale = 1, norm = 1, sparsity = 0.01, window = hann, scale = True, pad_mode = constant, res_type = soxr_hq, dtype = None
constant-Q chromagram	sr = 16 kHz, hop_length = 512, fmin = 32.7 Hz, norm = inf, threshold = 0.0, tuning = None, n_chroma = 12, n_octaves = 7, window = None, bins_per_octave = 36, cqt_mode = full

proper comparison. If emotion-level aggregation is required, filenames should include the relevant emotion label.

Once the input files are prepared, the evaluation script should be executed with the relevant directories specified as parameters, as illustrated in Listing 1. If the evaluation procedure is to consider emotional states and aggregate the results per emotion, the `--evaluate_emotion_transfer` flag should be included.

Listing 1. Command used to run the evaluation script.

```
python eval.py \
  --original_dir <directory_A> \
  --cloned_dir <directory_B>
```

The evaluation script generates two output files in the current directory – detailed metrics for each file pair and averaged results for the dataset, categorized by emotion if applicable.

VI. LEADERBOARD

The results of the evaluation process, using the *ClonEval* library, are presented in the Open Voice Cloning Leaderboard³ on Hugging Face (see Figure 2). The main page of the leaderboard consists of the aggregate results obtained by the models, which represent the average cosine similarity between the speaker embeddings of the reference and generated samples. These results were generated by the WavLM-Large model.

In addition to the overall results, two tabs are available for the Leaderboard:

- *Emotions*: Presents the averaged cosine similarity between the speaker embeddings obtained by the models on emotional speech datasets reported separately for each dataset and emotion and in total.
- *Features*: Presents the averaged cosine similarity between the values of selected acoustic features obtained by the models per dataset and overall.

³https://huggingface.co/spaces/amu-cai/Open_Voice_Cloning_Leaderboard

Open Voice Cloning Leaderboard

The **Open Voice Cloning Leaderboard** ranks and evaluates the voice cloning models across diverse datasets, including emotional speech. It also delivers an in-depth analysis of how different acoustic features shape the final results.

[Leaderboard](#) [About](#) [Submit here!](#)

Overall Emotions Features

The results represent the cosine similarity between the speaker embeddings of the original and cloned samples, generated by the WavLM model.

Rank	Model	Average	LS Clean	CREMA-D	RAVDESS	SAVEE	TESS
1	coqui/XTTS-v2	0.8356	0.8881	0.806	0.8539	0.8135	0.8167
2	microsoft/speecht5_vc	0.8298	0.9099	0.7618	0.8265	0.7987	0.8521
3	Plachtaa/VALL-E-X	0.7862	0.901	0.7412	0.7382	0.7674	0.7832
4	WhisperSpeech/WhisperSpeech	0.7837	0.9014	0.7284	0.6972	0.7725	0.8188
5	OuteAI/OuteTTS-0.2-500M	0.7499	0.8836	0.7359	0.7696	0.5394	0.8207

Citation

Fig. 2. Overview of the Open Voice Cloning Leaderboard on Hugging Face.

An additional tab, *Submit here*, has been included to simplify the process of submitting voice cloning models for evaluation by the community.

VII. EXPERIMENTS

A. Overall

Table II presents the average cosine similarity between the WavLM speaker embeddings of the reference and generated samples. The XTTS-v2 [12] achieved the best overall performance, while SpeechT5 [13] achieved the highest results on the LS test-clean and TESS datasets. Other models, including OuteTTS-0.2-500M [14], VALL-E X⁴ [15], and WhisperSpeech [16], produced slightly lower results. Across most datasets, the cosine similarity score exceeded 0.7, indicating effective voice cloning. The highest scores were observed for the LS test-clean dataset, reflecting better performance on non-emotional speech.

TABLE II

COSINE SIMILARITY BETWEEN SPEAKER EMBEDDINGS OF THE ORIGINAL AND CLONED SAMPLES GENERATED BY THE WAVLM MODEL.

Model	LS	CREMA	RAVD.	SAVEE	TESS	Avg.
OuteTTS	0.8836	0.7359	0.7696	0.5394	0.8207	0.7499
SpeechT5	0.9099	0.7618	0.8265	0.7987	0.8521	0.8298
VALL-E X	0.9010	0.7412	0.7382	0.7674	0.7832	0.7862
WhisperSpeech	0.9014	0.7284	0.6972	0.7725	0.8188	0.7837
XTTS-v2	0.8881	0.8060	0.8539	0.8135	0.8167	0.8356

B. Acoustic Features

Table III presents the average cosine similarity between the acoustic features extracted from the reference sample and those generated by each model. The OuteTTS model demonstrated the highest performance, though the results obtained by the other models were not significantly different in most cases.

High similarities for spectral-envelope and perceptual-scale features, such as LPCs, mel spectrograms, spectral contrast, and IIRT, are consistent with preservation of speaker-related characteristics. Lower similarities for MFCCs, spectrograms, and rhythm-related features reflect the intended difference in spoken content. Overall, these results support the WavLM-based speaker-similarity findings while showing that acoustic features capture different aspects of the generated speech.

C. Emotions

Table IV presents the average cosine similarity scores for each emotional state. For the CREMA-D dataset, XTTS-v2 achieved the highest average results across all emotions, with a slight advantage over other models. WhisperSpeech obtained the lowest average similarity and performed worst across all emotional states except for *fear* and *sadness*, where OuteTTS achieved the lowest results. The *neutral* state was cloned most successfully in all cases, while *fear* posed the greatest challenge. A similar trend was observed in the RAVDESS dataset. XTTS-v2 achieved the highest average results, and WhisperSpeech achieved the lowest. The *neutral* state was

⁴An open-source implementation was used. Available at <https://github.com/Plachtaa/VALL-E-X>

TABLE III
COSINE SIMILARITY BETWEEN THE VALUES OF SELECTED ACOUSTIC FEATURES OF THE ORIGINAL AND CLONED SAMPLES OBTAINED FOR LS TEST-CLEAN DATASET.

Feature	OuteTTS	SpeechT5	VALL-E X	WhisperSpeech	XTTS-v2
pitch	0.6094	0.5278	0.5818	0.5863	0.5287
spectrogram	0.2609	0.2406	0.2451	0.2316	0.2298
mel spectrogram	0.9259	0.9109	0.9208	0.9091	0.8622
MFCCs	0.0486	0.0189	0.0494	0.0261	0.0352
RMS	0.6970	0.6040	0.6810	0.6400	0.6238
spectral centroid	0.7674	0.7855	0.7608	0.7485	0.7387
spectral bandwidth	0.9318	0.9379	0.9256	0.9238	0.9048
spectral contrast	0.9365	0.9423	0.9381	0.9359	0.9392
spectral flatness	0.3229	0.3091	0.3199	0.2655	0.2418
spectral roll-off	0.8198	0.8357	0.8134	0.8037	0.7907
zero-crossing rate	0.6115	0.6052	0.6057	0.5913	0.5785
LPCs	0.9720	0.9760	0.9747	0.9628	0.9808
tempogram	0.5159	0.5127	0.5030	0.5071	0.3658
chromagram	0.6036	0.6312	0.5966	0.5694	0.5775
pseudo-constant-Q transform	0.6707	0.6261	0.6499	0.6388	0.6242
IIRT	0.9550	0.9500	0.9559	0.9531	0.9520
variable-Q transform	0.9002	0.8910	0.8991	0.8881	0.8787
constant-Q chromagram	0.7167	0.7330	0.7146	0.6836	0.7138

once again the easiest to clone, and *fear* was the hardest. For the SAVEE dataset, XTTS-v2 demonstrated the highest overall average performance and achieved the best results for all emotional states except *disgust* and *fear*, for which SpeechT5 performed best. The OuteTTS model produced the poorest results for all emotions. Contrary to other datasets, *fear* was cloned most successfully, while *disgust* was the most challenging emotion. The results were slightly different for the TESS dataset. SpeechT5 achieved the highest average results, and VALL-E X obtained the lowest. OuteTTS performed best in cloning *happiness*, while XTTS-v2 outperformed the others in cloning *sadness*. As with CREMA-D and RAVDESS, the highest average similarities were obtained for the *neutral* state, while *fear* was the most difficult to replicate. Overall, the XTTS-v2 model achieved the best results for all emotions except *fear*, but it is worth noting that the similarity for this emotion differed from the best by 0.0067. The OuteTTS model performed worst on all emotions. It is worth noting that all models performed best at cloning the neutral state, which is not surprising given the narrative nature of this emotion. Three of the models – OuteTTS, VALL-E X and XTTS-v2 – performed worst at cloning the emotional state of *fear*. The emotion of *anger* was the biggest problem for SpeechT5 and for WhisperSpeech it was *disgust*.

VIII. CONCLUSION

We introduced *ClonEval*, a standardized and reproducible benchmark for evaluating voice cloning systems to address the growing need for automatic, reliable and architecture-agnostic evaluation methods. By combining a deterministic evaluation protocol, an open-source software library, and a public leaderboard, *ClonEval* provides a comprehensive framework for assessing voice cloning models. Our benchmark builds upon widely recognized evaluation techniques while introducing well-defined defaults to minimize ambiguity and enhance reproducibility.

The preliminary results reported in Section VII show that the performance of the models under study varies when we consider individual emotions. While all the models performed best when cloning emotionally neutral utterances, their behavior differed with respect to *fear*, *anger* and *disgust*. The majority of the evaluated models under-performed while cloning utterances that included *fear*.

As part of our future work, we plan to expand our evaluation framework by applying it to a wider variety of voice cloning models and diverse datasets. Additionally, we plan to incorporate human evaluation to complement objective metrics with subjective judgments. This will provide insights into the perceived quality and naturalness of the generated audio samples.

IX. LIMITATIONS

We acknowledge that this work has several limitations. First, there is a potential risk of data contamination, as some of the evaluated models may have been trained on the same datasets used for evaluation. This could potentially inflate the obtained results. Second, the scope of our evaluation of emotional speech is limited by the availability of existing datasets. For instance, the TESS dataset includes only two speakers, and the RAVDESS dataset contains only two distinct utterances. Finally, although WavLM is widely adopted for measuring similarity between audio samples, the extent to which its scores align with human judgment remains unclear.

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TABLE IV
COSINE SIMILARITY BETWEEN SPEAKER EMBEDDINGS OF THE ORIGINAL AND CLONED SAMPLES GENERATED BY THE WAVLM MODEL FOR EMOTIONAL STATES.

Dataset	No. samples	Emotion	OuteTTS	SpeechT5	VALL-E X	WhisperSpeech	XTTS-v2
CREMA-D	7 442	anger	0.7613	0.7834	0.7488	0.7305	0.7962
		disgust	0.7428	0.7522	0.7388	0.7272	0.8091
		fear	0.7046	0.7409	0.7207	0.7246	0.8052
		happiness	0.7370	0.7612	0.7457	0.7201	0.7969
		neutral	0.7637	0.7875	0.7671	0.7459	0.8241
		sadness	0.7101	0.7493	0.7296	0.7249	0.8072
		Average	0.7366	0.7624	0.7418	0.7289	0.8065
RAVDESS	1 440	anger	0.7815	0.8128	0.7683	0.6610	0.8459
		disgust	0.7588	0.8273	0.7430	0.6936	0.8581
		fear	0.7388	0.7737	0.7552	0.6773	0.8143
		happiness	0.8004	0.8361	0.7670	0.6706	0.8580
		neutral	0.7911	0.8697	0.7409	0.7583	0.8815
		sadness	0.7630	0.8383	0.7316	0.7157	0.8554
		Average	0.7723	0.8263	0.7510	0.6961	0.8522
SAVEE	480	anger	0.5696	0.7483	0.7645	0.7899	0.8201
		disgust	0.4717	0.8114	0.7541	0.7579	0.7990
		fear	0.5965	0.8446	0.7868	0.8087	0.8160
		happiness	0.5436	0.7968	0.7603	0.7872	0.8102
		neutral	0.5591	0.7835	0.7876	0.7228	0.8224
		sadness	0.5291	0.8001	0.7629	0.7650	0.8211
		Average	0.5449	0.7975	0.7694	0.7719	0.8148
TESS	2 800	anger	0.7664	0.8247	0.7678	0.8034	0.7770
		disgust	0.8403	0.8781	0.8040	0.8044	0.8637
		fear	0.7413	0.8392	0.7238	0.8300	0.7362
		happiness	0.8506	0.8329	0.7900	0.8069	0.7989
		neutral	0.8343	0.8882	0.7842	0.8587	0.8641
		sadness	0.8519	0.8520	0.7860	0.8008	0.8622
		Average	0.8141	0.8525	0.7760	0.8174	0.8170
Average		anger	0.7197	0.7923	0.7623	0.7462	0.8098
		disgust	0.7034	0.8172	0.7600	0.7458	0.8325
		fear	0.6953	0.7996	0.7466	0.7601	0.7929
		happiness	0.7329	0.8068	0.7658	0.7462	0.8160
		neutral	0.7370	0.8322	0.7699	0.7714	0.8480
		sadness	0.7135	0.8099	0.7525	0.7516	0.8365
		Average	0.7170	0.8097	0.7595	0.7536	0.8226

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